Count groups of wiggles

Why Electronics Use Sine Waves

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little draws 2 groups of swing marks, with 3 marks in each group. How many marks are there?

3 + 3 = _____

Show your thinking.

2. Big draws spring bounces in 4 rows, with 2 bounces in each row. How many bounces are drawn?

2 + 2 + 2 + 2 = _____

Show your thinking.

3. Little counts 3 sets of rope wiggles, with 3 wiggles in each set. How many wiggles is that?

3 + 3 + 3 = _____

Show your thinking.

Count groups of wiggles

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4. A page has 3 rows of wave drawings, with 4 drawings in each row. How many drawings are on the page?

$4 + 4 + 4 =$ _____

Show your thinking.

5. Big counts 5 groups of crank turns, with 3 turns in each group. How many turns does Big count?

$3 + 3 + 3 + 3 + 3 =$ _____

Show your thinking.

6. Little draws speaker wiggles in 4 rows, with 5 wiggles in each row. How many wiggles are drawn?

$5 + 5 + 5 + 5 =$ _____

Show your thinking.

A smooth wiggle

Little drew a smooth line that rose and fell again. Big called it a sine wave. Could this shape help explain a speaker?

Slow at the turn

The swing slowed at each end and sped through the middle. A sine-like line could record its back-and-forth motion as time passed.

A bouncing spring

Big held a spring above a tray. Its end bounced up and down. Each bounce grew smaller until the spring settled.

Along the rope

Little moved a rope from side to side. A wiggle traveled toward Big, while each bit of rope moved back and forth.

Turning into power

A generator turns motion into electrical energy. As a magnet spins beside wire, the electric push grows, fades, and changes direction smoothly.

A speaker's rhythm

The speaker's cone moved in and out to make a steady tone. Little drew its smooth rhythm. Where else might a wiggle appear?

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

Why Electronics Use Sine Waves

Give a wiggle its words

Why Electronics Use Sine Waves · Curiosity day

Name _____ Date _____

Say one clear command, then draw or write it.



Slow at the turn



A bounce comes back



A wiggle travels

Draw a swing, a spring, or a rope. Show which way it moves.

sine wave · wiggle · swing · spring · smooth · gently · back · forth

Swing … tell how to move gently

Watch … tell what to notice about a bounce

Draw … describe the smooth wiggle

Does your command tell what to do, using a word such as Swing, Watch, or Draw?

Why Electronics Use Sine Waves

01 What is a sine wave?



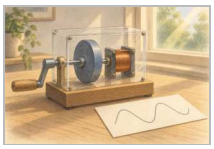
A gentle swing slows near each turn.

Look closer. Notice how the swing slows near the end of its trip.

A little explanation. A sine wave is a smooth back-and-forth wiggle that repeats. It can show how a position or an electric push changes over time. A gentle swing moves in a sine-like rhythm; its seat travels along an arc.

Try this. With Big nearby, take a gentle swing, bounce a soft toy spring over a tray, or move a rope from side to side on the floor. Draw the smooth wiggle you notice. Keep wall power and plugged-in devices closed and untouched; these are activities with a swing, spring, rope, paper, or an intact battery speaker.

02 How can spinning make a wave?



Spinning can make a smoothly changing electric push.

Look closer. Find the turning disk beside the wire coil.

A little explanation. A generator turns motion into electrical energy. As a magnet spins beside wire, the electric push changes smoothly, first one way and then the other. Many generators make the AC power supplied at home. AC means that the push changes direction repeatedly.

Try this. Draw a dot on a paper circle. Turn the circle slowly and watch the dot rise and fall.

03 What wiggles inside a speaker?



A speaker moves air to make sound.

Look closer. Look at the circle on the front of the speaker.

A little explanation. The speaker cone moves in and out, making the air nearby squeeze and spread. Those changes travel to your ears as sound. For a pure, steady tone, the cone moves in a sine-like rhythm. Music and voices make more complicated wiggles.

Try this. Let Big set an intact battery speaker to low volume. Rest fingertips lightly on its outer case, leaving the cone alone, and notice any quiet buzz. A drawing of the cone's motion works too.

04 Why use a smooth wiggle?



A sine wave is a smooth, repeating wiggle.

Look closer. Follow the rounded turns in Little's notebook.

A little explanation. Spinning parts can make sine waves naturally. Smooth changes help many motors and speakers work evenly. Sudden jumps can make extra buzz or jerky starts. Electronics also use steady power and other signal shapes; not every signal is a sine wave.

Try this. Draw a smooth wave and a zigzag. Trace them slowly with a finger and compare the turns.